Andrew Zagula

(908) 392-6724 | andrewzagula800@gmail.com | github.com/andrewzagula | andrewzagula.com

Education

California Institute of Technology Pasadena, CA
B.S. in Computer Science, Minors in Robotics *Expected Graduation, Jun 2029*

Experience

FinalDose (YC P26) Remote
Software Engineer *Aug 2026 – Present*

- Built LLM-agnostic 36-tool MCP server and 22 REST endpoints over shared Biopython engine for DNA parsing, alignment, and lab-procedure simulation; shipped AI-drafted lab records and notes for scientist review and sign-off
- Engineered lab data platform (Django, RDS/PostgreSQL, TypeScript/Next.js) on AWS ECS via Docker/Terraform and company-domain Google SSO; integrated Drive API for editable documents and S3 for immutable records

Terranox AI (YC W26) San Francisco, CA
Software Engineer *Jun 2026 – Aug 2026*

- Shipped land evaluation dashboard (TypeScript/Next.js) over 8,000 parcels, spatially querying a 200K-file AWS RDS/PostGIS index over S3-backed survey data across table, 2D map, 3D layer-stack, and heat map grid views
- Designed LLM scoring pipeline (Python) ranking 8,000 land parcels by owner sell likelihood, parsing 150 public companies' filing PDFs into normalized weighted scores over cash runway, owner type, and resale history

University of California, Berkeley Remote
Student Researcher *Jul 2024 – Dec 2025*

- First-authored AutoAdv (NeurIPS Lock-LLM 2025), an automated adversarial prompting framework for multi-turn LLM jailbreaking; achieved 99% attack success rate against Qwen3-235B in 6 turns, +24% vs. single-turn (75%)
- Implemented two-phase prompt rewriting pipeline with attacker LLM Grok 3 Mini and GPT-4o mini judge; developed 4 sampling temperature strategies and cataloged 28 effective rewriting techniques to guide future attacks

Boston University Boston, MA
Research Intern (RISE) *Jun 2025 – Aug 2025*

- Co-authored BabyVLM-V2 (CVPR 2026), a developmentally inspired framework for sample-efficient vision-language models; constructed DevCV Toolbox, a 10-task benchmark generated from 478 hours of infant-perspective video data
- Developed OpenCV pipeline with frame sampling and sliding-window segmentation; applied YOLOE-11 for object detection and tracking, filtering 2.2K+ usable clips (∼63% yield) from 3.5K+ candidates via Label Studio

Projects

PaperTrail | *Python, TypeScript, FastAPI, Next.js, LangGraph, OpenAI API, ChromaDB, SQLite*

- Built research assistant with arXiv/PDF ingestion, LLM-generated multi-query paper discovery, structured paper breakdowns, and 3 LangGraph workflows (comparison, ideation, and implementation planning) with retry cycles
- Developed provider-agnostic LLM layer with schema-validated output and self-repair across 11 workflow steps; built RAG pipeline for multi-turn Q&A with section-level citations over ChromaDB top-5 retrieval and SQLite

AUDIT | *Python, Typer, OpenAI API, ChromaDB, SQLite, PyInstaller, npm*

- Built npm-installable CLI vulnerability scanner producing scored JSON reports from AST/regex chunking, heuristic prefiltering, and parallel LLM review via ChromaDB retrieval over 50-pattern CWE/OWASP knowledge base
- Engineered route analysis (Express/FastAPI/Flask) with 3-hop import resolution and 8-level router nesting; caught multi-file flaws and unauthenticated admin routes at 96%+ recall on 55 planted vulnerabilities (<2% false positives)

Technical Skills

Languages: Python, TypeScript, JavaScript, C++, Java, SQL
Frameworks: React, Next.js, Node.js, FastAPI, Django, Tailwind CSS
Infrastructure: AWS, Docker, Terraform, CI/CD (GitHub Actions), PostgreSQL/PostGIS, Supabase, Firebase, Redis
Libraries/Tools: PyTorch, Hugging Face, LangGraph, MCP, OpenAI/Anthropic APIs, OpenCV, ChromaDB, SQLite

Awards

- 2x USAMO Qualifier, 5x AIME Qualifier, USA Computing Olympiad Gold, NY Tech Week $1K Hackathon Winner